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    hand_eye_diff_render_comparison.py — markerless hand-eye calibration by differentiable rendering
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Dataset   : /home/aseshad4/hand_eye_collection_scripts/gt_dataset  (12 frames; tip found in 12, 7 kept after outlier rejection)
Generated : 2026-10-26T18:32:19   fit runtime 132s
Ground truth = marker registration + 3D pose alignment (cam_T_base_post_alignment); used ONLY for scoring.
DR inputs   = rectified right+left images, intrinsics / rect_T_cam_raw / stereo P, joint.stl, PSM kinematics.
              No marker calibration, no ground truth, no initial pose, no annotation.
Pipeline    = stereo tool box -> GrabCut -> PnP init -> 4 DR passes (masks re-seeded from the
              DR estimate after pass 1) -> outlier rejection + refit.

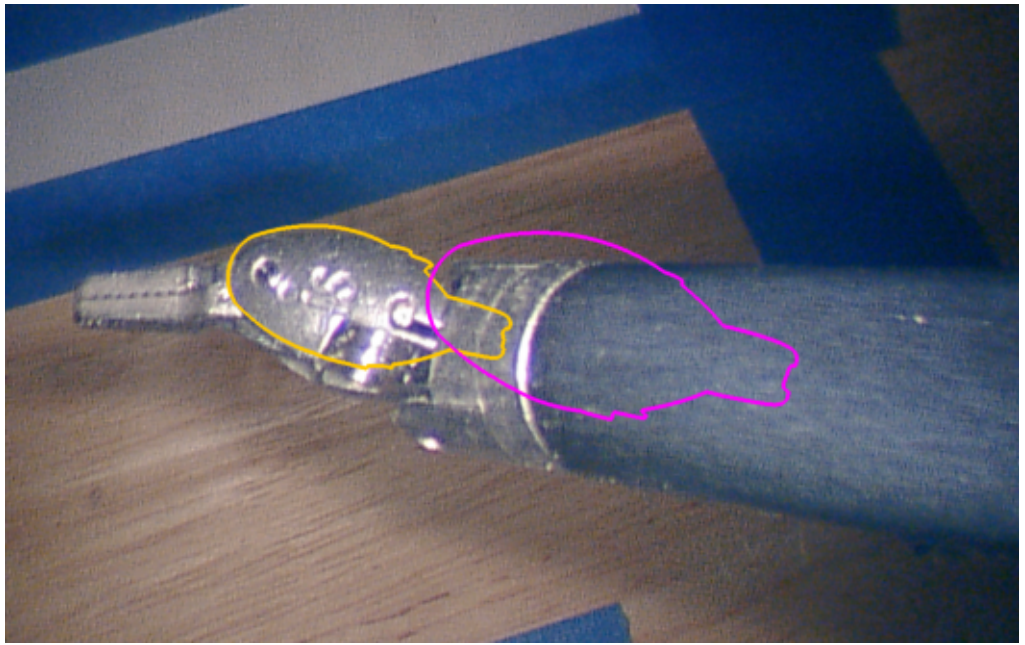
Whole calibration vs ground truth (all frames):
                            clevis displ. (mm)  rotation (deg)  base origin (mm)
  PnP init (no prior)                    26.63           17.57             40.09
  marker only                            19.13            0.00             19.13
  diff-render (markerless)                4.43            6.91             10.92

First 5 frames (clevis pose vs its pose under the ground truth; IoU of the rendered silhouettes):
  frame kept       marker only t(mm)  rot(deg)    IoU   diff-render (m t(mm)  rot(deg)    IoU
  ------------------------------------------------------------------------------------------
      0    y                   19.13      0.00  0.056                   4.87      6.91  0.783
      1    n                   19.13      0.00  0.000                   4.66      6.91  0.753
      2    y                   19.13      0.00  0.002                   4.26      6.91  0.737
      3    y                   19.13      0.00  0.008                   4.45      6.91  0.678
      4    y                   19.13      0.00  0.000                   4.62      6.91  0.673
  ------------------------------------------------------------------------------------------
   mean                       19.13      0.00  0.013                   4.57      6.91  0.725
  (t = clevis origin / pitch pin; rotation is the calibration's rotation error, the same for every frame)

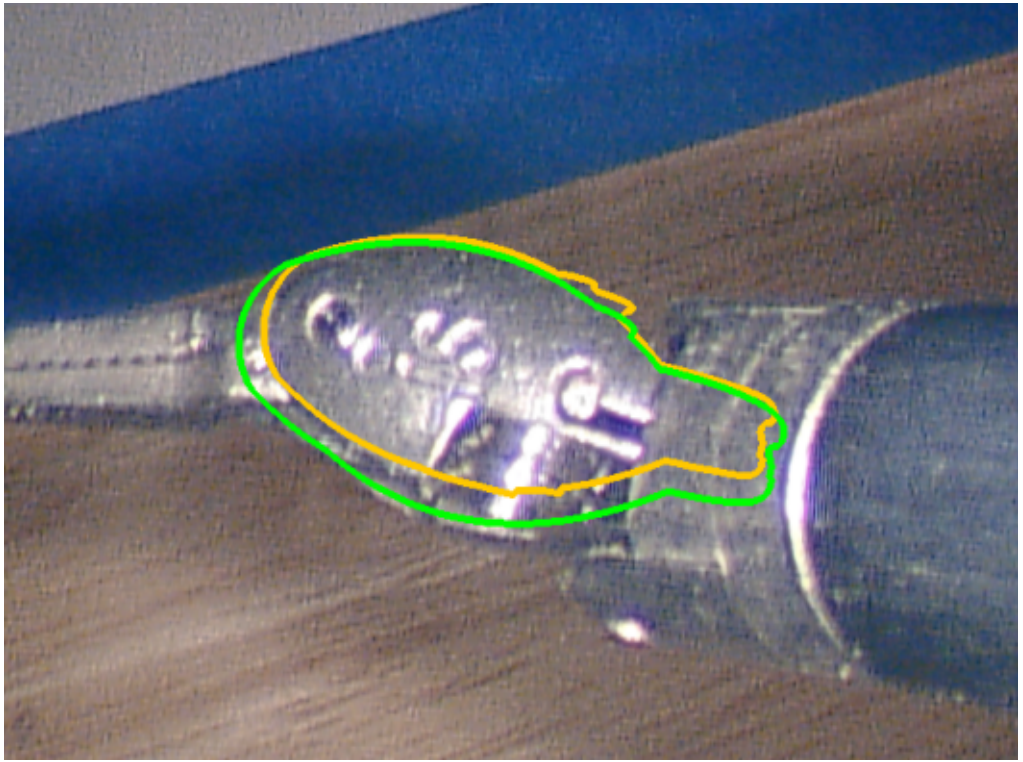
cam_raw_T_base (m), diff-render:
  [ 0.434501 -0.213425  0.87502   0.051076]
  [-0.246257  0.90635   0.343348 -0.003901]
  [-0.866352 -0.364665  0.341252 -0.015199]
  [0. 0. 0. 1.]
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gt_dataset frame 0: clevis rendered at each calibration (coloured) and at the ground truth (orange)

marker only vs ground truth
translation error 19.13 mm rotation error 0.00 deg IoU 0.056

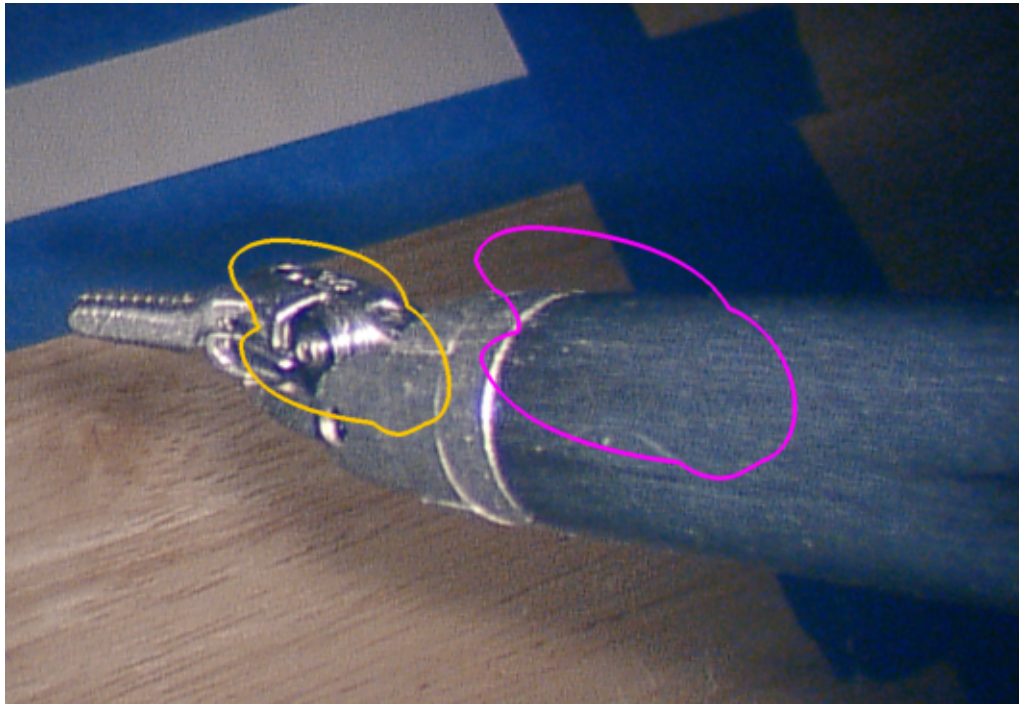


diff-render (markerless) vs ground truth
translation error 4.87 mm rotation error 6.91 deg IoU 0.783

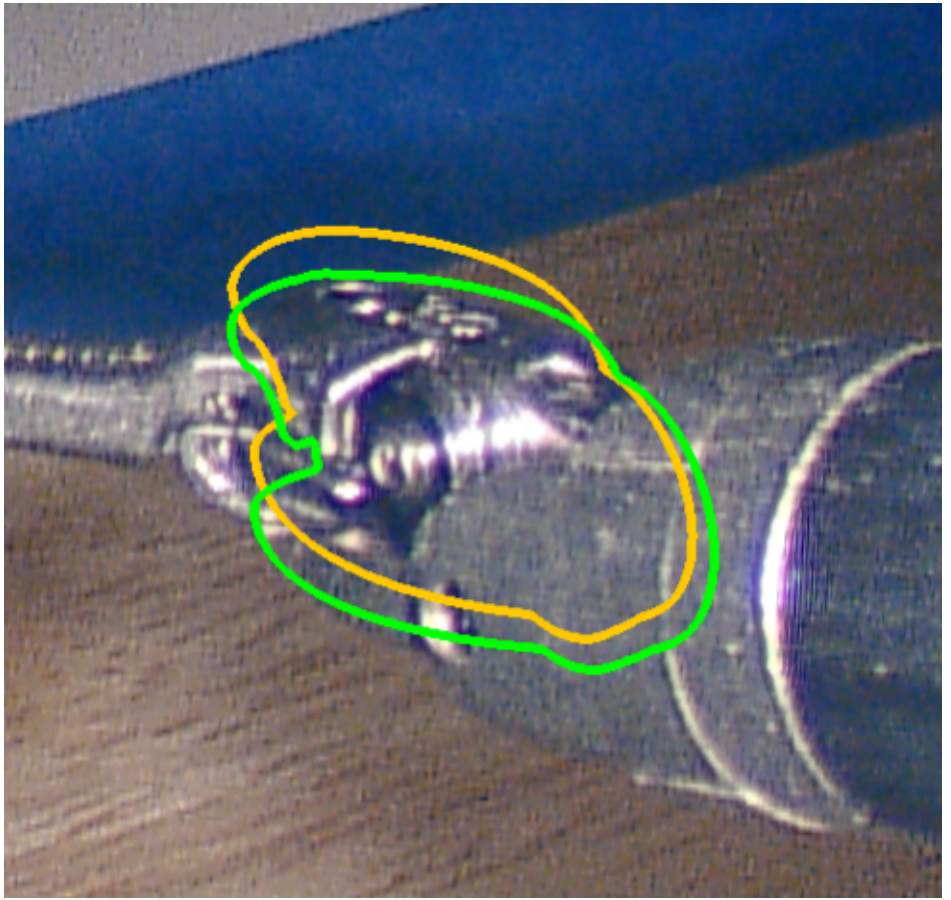


gt_dataset frame 1: clevis rendered at each calibration (coloured) and at the ground truth (orange) [frame not used by the DR fit]

marker only vs ground truth
translation error 19.13 mm rotation error 0.00 deg IoU 0.000

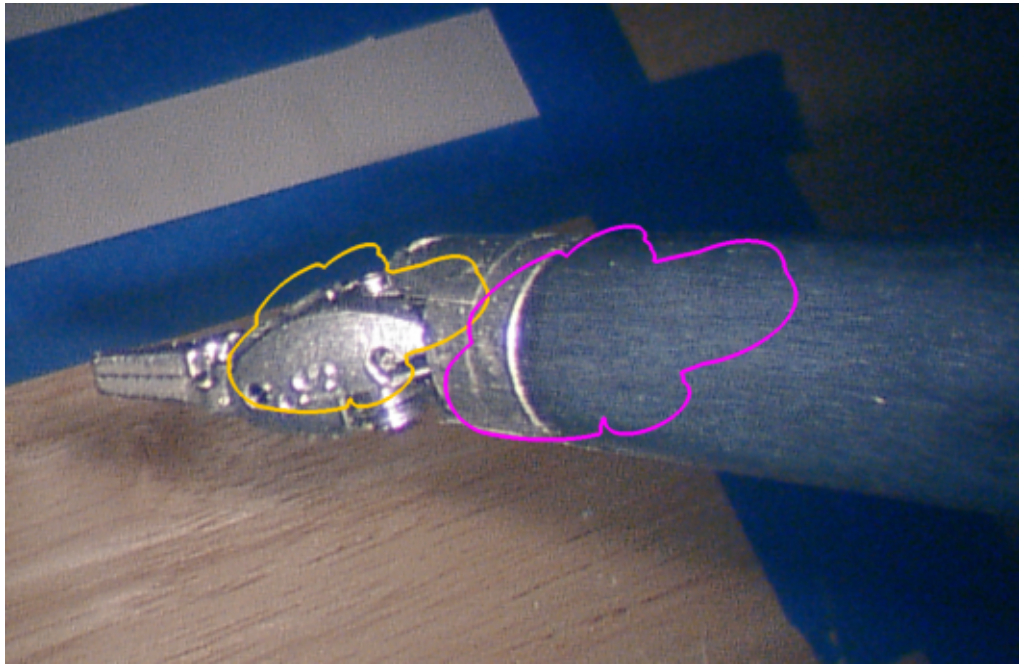


diff-render (markerless) vs ground truth
translation error 4.66 mm rotation error 6.91 deg IoU 0.753

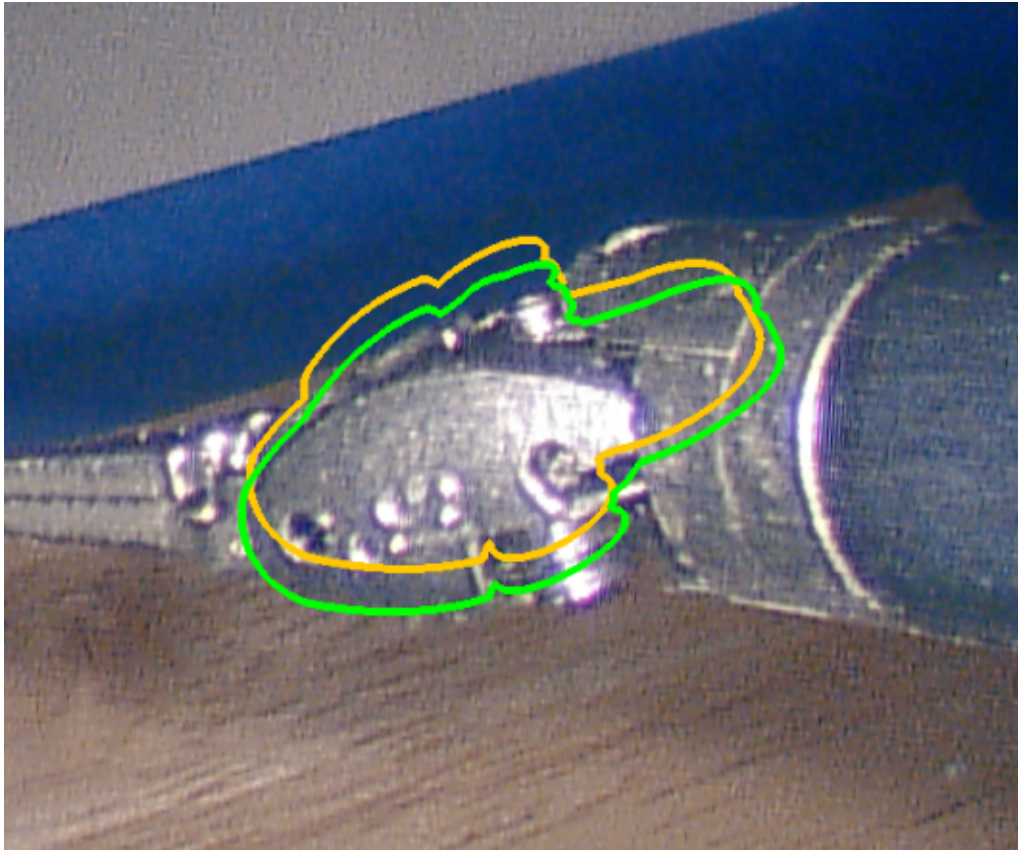


gt_dataset frame 2: clevis rendered at each calibration (coloured) and at the ground truth (orange)

marker only vs ground truth
translation error 19.13 mm rotation error 0.00 deg IoU 0.002

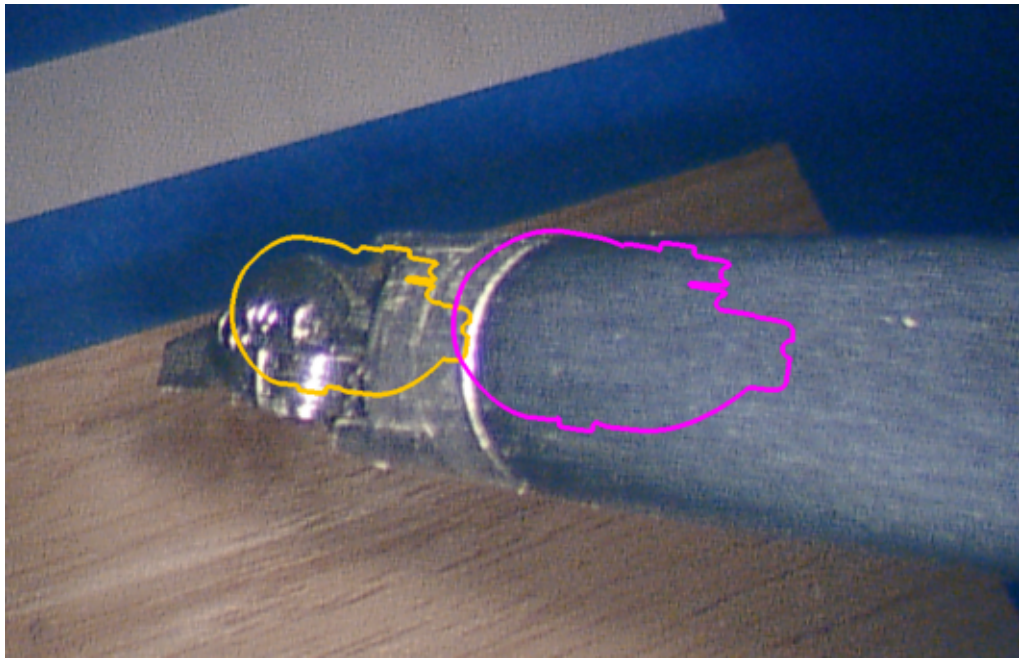


diff-render (markerless) vs ground truth
translation error 4.26 mm rotation error 6.91 deg IoU 0.737

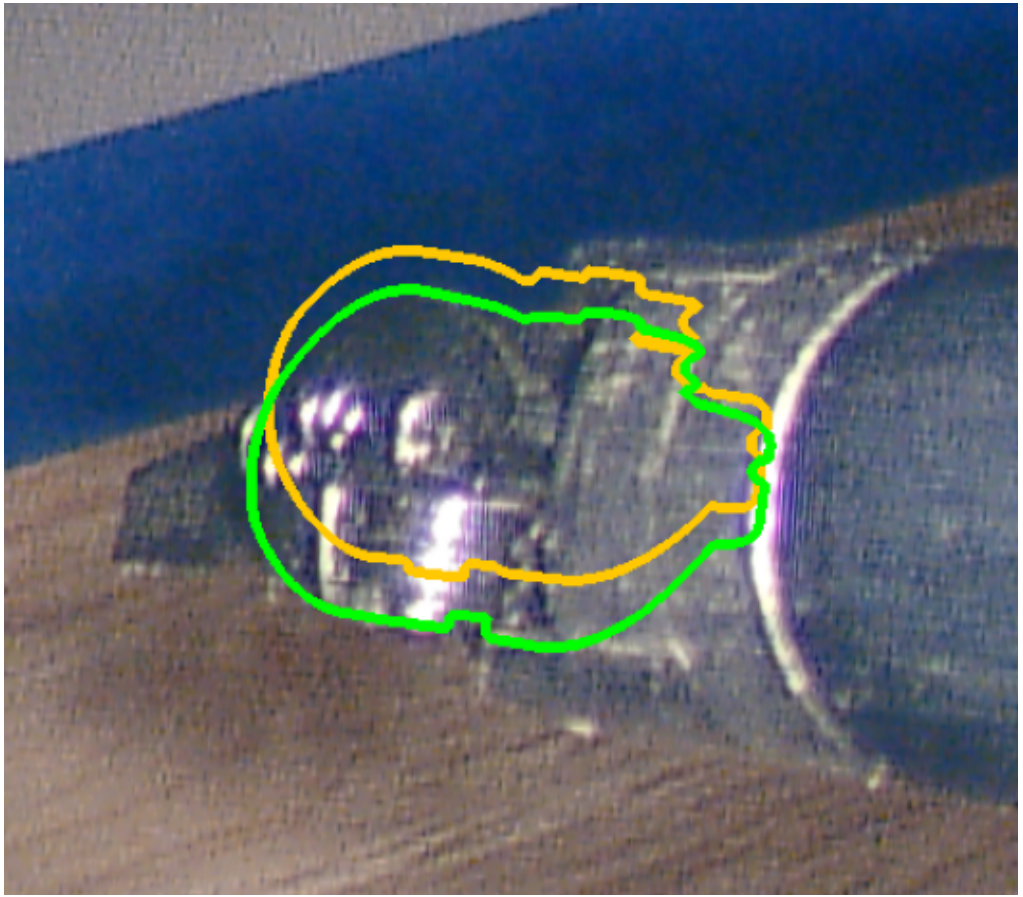


gt_dataset frame 3: clevis rendered at each calibration (coloured) and at the ground truth (orange)

marker only vs ground truth
translation error 19.13 mm   rotation error 0.00 deg   IoU 0.008

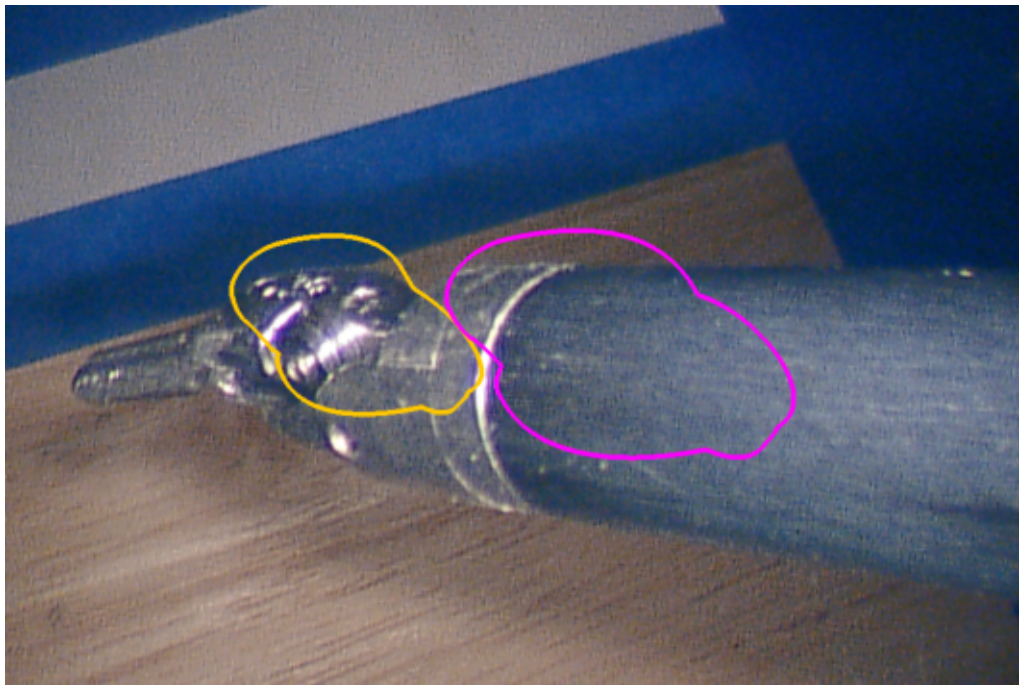


diff-render (markerless) vs ground truth
translation error 4.45 mm   rotation error 6.91 deg   IoU 0.678



gt_dataset frame 4: clevis rendered at each calibration (coloured) and at the ground truth (orange)

marker only vs ground truth
translation error 19.13 mm rotation error 0.00 deg IoU 0.000



diff-render (markerless) vs ground truth
translation error 4.62 mm rotation error 6.91 deg IoU 0.673

